



RV College of
Engineering®

Traffic Monitoring System Using **YOLO** and **Deep SORT**



RV College of
Engineering®

Chandan Kumar K R	1RV23CS402
Gopinath Ramaje	1RV23CS404
KIRAN H R	1RV23CS405

Introduction

- The rapid growth of urbanization has led to a significant increase in vehicular traffic, making efficient traffic monitoring essential for ensuring road safety and effective traffic management.
- This project aims to develop a real-time traffic monitoring system that utilises advanced object detection and tracking algorithms, namely YOLO (You Only Look Once) and Deep SORT (Simple Online and Realtime Tracking).

Last Semester's Project Overview

- Object Detection and Tracking using YOLOv8 & Deep SORT.
- **Objective:** Track multiple objects in real-time with high accuracy.
- **Key Challenges Addressed:**
 - Handling occlusions and fast-moving objects.
 - Reducing identity switches.
 - Improving tracking accuracy.

System Architecture

1

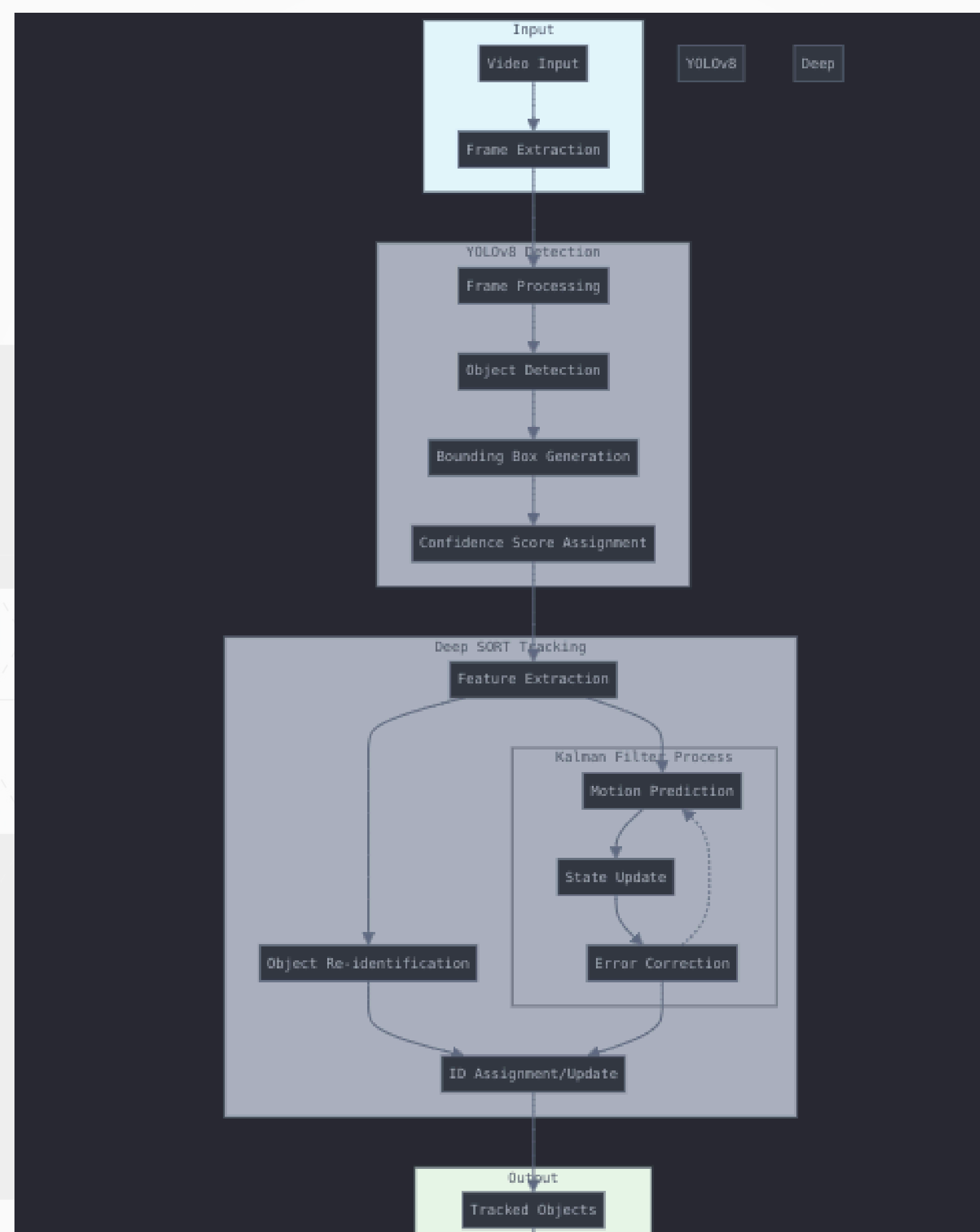
YOLO for object detection:
Frames extracted & objects
detected

2

Deep SORT for tracking

3

Kalman Filter:
for Motion Estimation



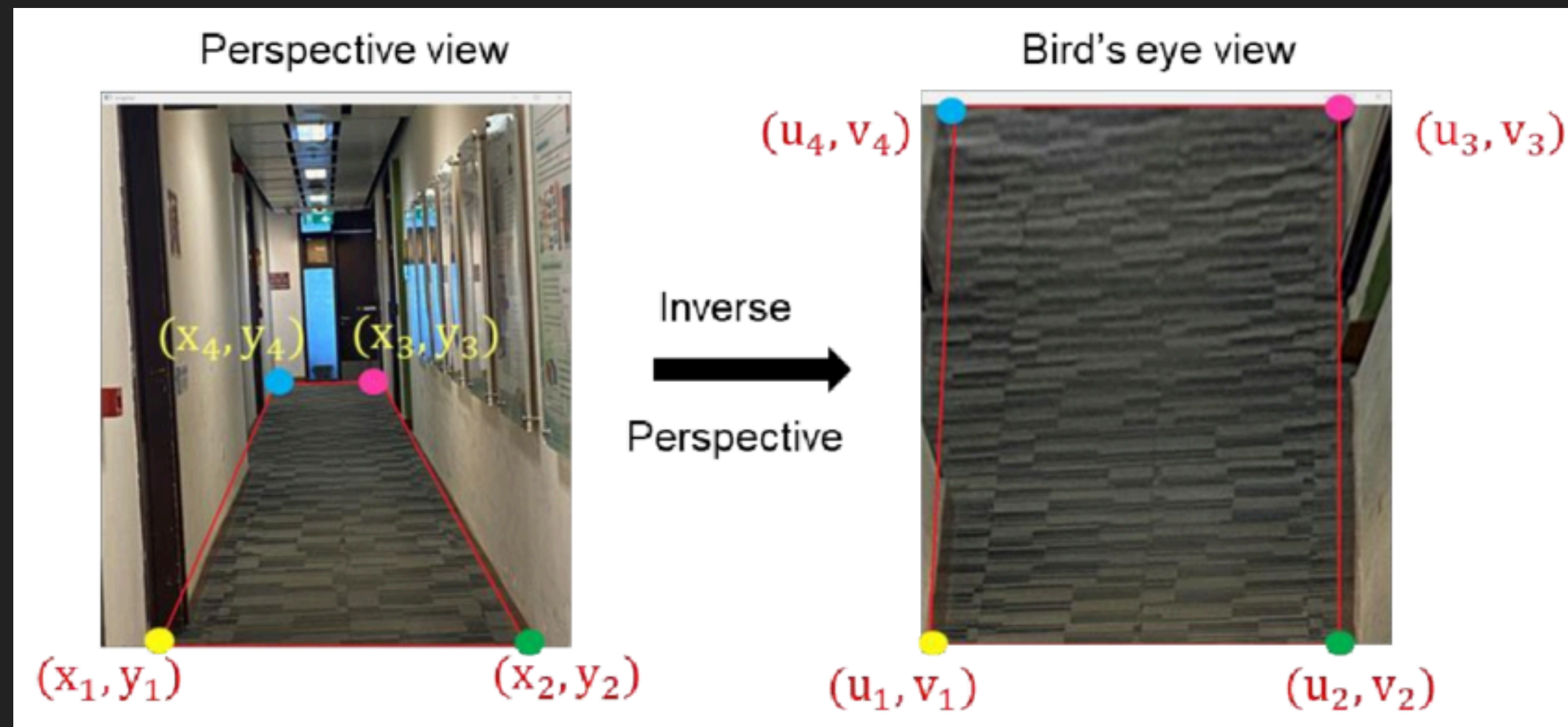
Transition to Traffic Monitoring Application

- The shift from general object tracking to traffic-specific monitoring.
- Integration with additional modules like speed estimation and ticketing .

Bird's Eye View Transformation

- When tracking vehicles on the road using a normal camera, perspective distortion makes it difficult to measure distances accurately. Vehicles closer to the camera appear larger, and those further away appear smaller.
- To solve this, we use a Bird's Eye View (BEV) transformation, also called perspective transformation, which converts the road scene from a camera perspective to a top-down view.

Bird's Eye View Transformation



Speed Calculation:

1. Measuring Pixel-Based Distance

- Once a vehicle's bottom-center position is mapped to the bird's eye view, we store its y-coordinates over time in a dictionary.
- The difference between consecutive y-coordinates gives the pixel distance traveled.
- Larger movements in consecutive frames indicate higher speed.

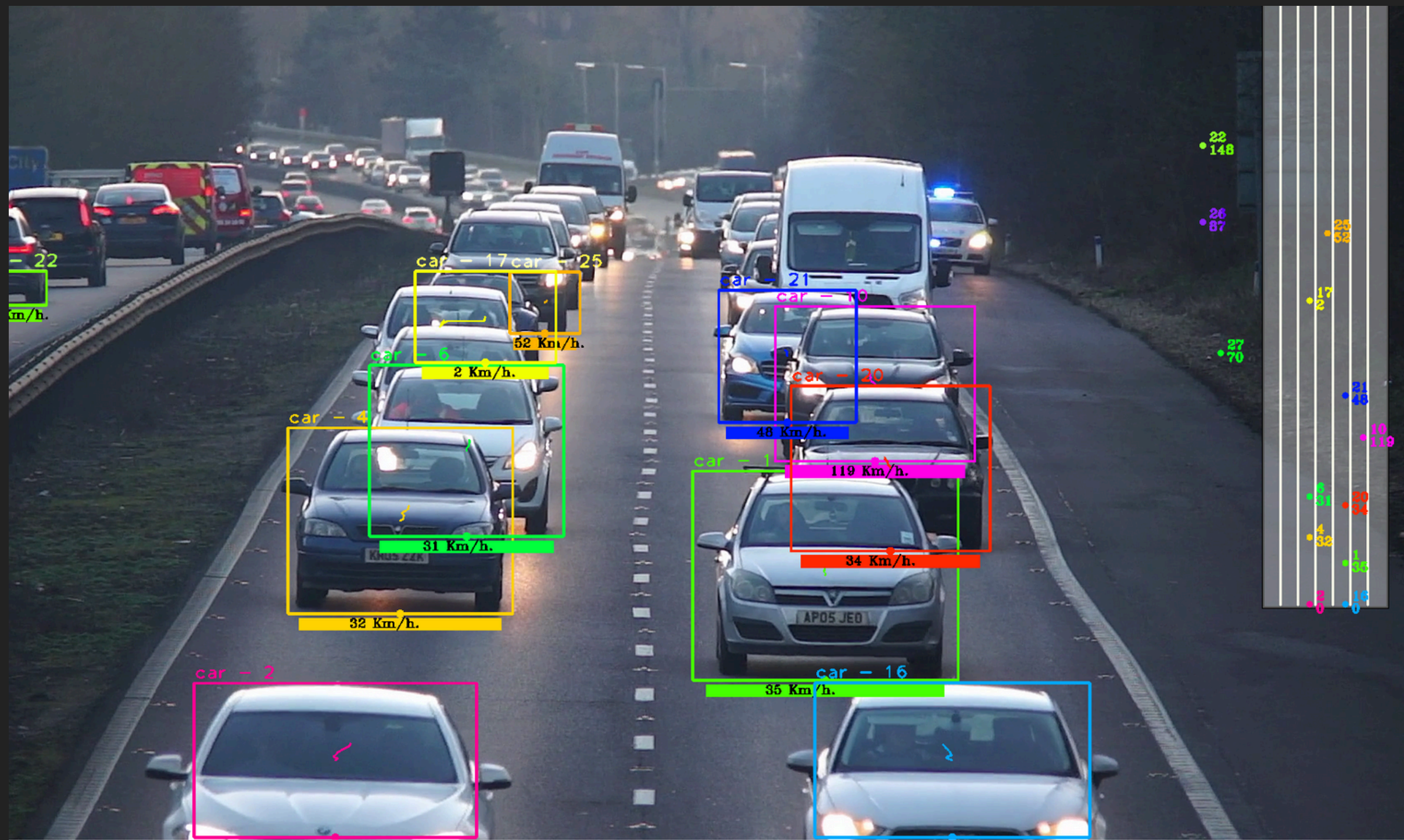
Speed Calculation:

2. Converting Pixel Distance to km/h

- To convert the pixel movement into real-world speed, we use a scaling factor based on a known reference distance in the transformed frame.
- The formula for speed calculation is:

$$\text{Speed (Km/h)} = \left(\frac{\text{Pixel Distance}}{\text{Pixels per Meter}} \right) \times \frac{3600}{\text{FPS} \times 1000}$$

Output:



The background is a solid green color with a pattern of white-outlined hexagons of various sizes. Some hexagons are solid white, while others are just outlines. The text 'Thank You' is centered within a white rectangular border.

Thank
You